

TRIAGELOOP.AI

A Safety-Oriented, Deadline-Aware Control Loop for Emergency-Department Waiting Rooms

Technical Working Paper and Reproducibility Monograph

Patient need becomes time. Time is tested against capacity. A clinician closes the loop.

Prepared for: Accenture Innovation Challenge 2026 - Round 2
Problem track: PatientTriage.ai
System version: Prototype 0.4.0 / Model 2.0.0 / Simulation 1.1.0
Evidence lock: 22 August 2026
Document status: Technical release candidate
Authors: TriageLoop Team

Synthetic decision-support prototype - not validated for clinical use.

This document reports engineering and synthetic-simulation evidence. It does not establish clinical effectiveness, patient benefit, safe staffing, legal compliance, or production readiness.

Abstract

Emergency-department triage is usually represented as a one-time severity classification made at arrival. The real waiting-room safety problem is temporal: a patient's condition and the department's ability to respond can both change after that classification. TriageLoop is a nurse-led research prototype that converts repeated observations into a conservative patient-specific *Dynamic Action Window*, estimates whether the live queue can meet that window, and exposes the difference as *Clinical Slack*. Its quantitative path combines age-aware deterministic rules, calibrated discrete-time trajectory risk, class-conditional conformal prediction, out-of-distribution checks, a discrete-event Queue Twin, and a bounded Next Best Observation mechanism. A clinician may accept, modify, or override every recommendation; material events are stored in a recomputed SHA-256 hash chain.

The work was developed under a predeclared evaluation protocol using 10,000 seeded synthetic longitudinal encounters, 30,020 time-indexed snapshots, 28 curated edge cases, three site profiles, and paired queue simulations. The selected compact boosted-hazard model passed the registered synthetic recall, calibration, and critical-class coverage gates. Across 1,200 policy shifts under simulated 3x load, TriageLoop reduced Action Window misses by 20.5% (95% CI 17.4%-23.8%) versus fixed 15-minute periodic re-triage. The effect remained directionally positive at 10-, 20-, and 30-minute response definitions, although only the 30-minute definition exceeded 20%. The single-observation hypothesis failed: it requested 87.5% fewer observations but lost 7.1 percentage points of operational critical recall. The product consequently releases Next Best Observation only as a first step, with full reassessment or escalation as the required fallback. An open MIMIC-IV demo supplied a bounded input-scale plausibility check over 63,163 measurements; it is not ED or clinical validation.

Two independent Claude review cycles were handled as red-team inputs rather than authority. Verified findings produced a stronger comparator, explicit alert-workload reporting, corrected explanation attribution, response-window sensitivity, six additional product surfaces, and a retained Docker clean-machine release gate. The final prototype passed 73 automated tests, TypeScript verification, an optimized production build, and a browser-verified deterioration-to-audit journey. The contribution is not a claim to have invented AI triage or dynamic queues; it is a safety-oriented recomposition that keeps clinical need, uncertainty, operational feasibility, human authority, and evidentiary limitations visible in one control loop.

Keywords: emergency-department triage; clinical deterioration; conformal prediction; discrete-time hazard; queue simulation; deadline scheduling; human-in-the-loop; uncertainty; safety engineering; synthetic evaluation.

Contents

1	Executive Technical Synopsis	5
1.1	Evidence snapshot	8
2	Problem Definition and Official-Brief Coverage	8
2.1	The waiting-room failure mode	8
2.2	Requirement-to-mechanism traceability	9
3	Research Lineage and Differentiation	10
3.1	Clinical and machine-learning foundations	10
3.2	Cross-domain transfer	10
3.3	Prior art and novelty boundary	11
4	Safety Contract and System Invariants	11
5	Data Strategy and Synthetic Longitudinal Cohort	12
5.1	Why synthetic data is primary	12
5.2	Population and split	13
5.3	Observation contract and temporal leakage controls	13
5.4	Curated cases	15
6	Trajectory-Risk and Uncertainty Methodology	15
6.1	Prediction target	15
6.2	Candidates and model selection	15
6.3	Asymmetric decision cost	16
6.4	Conformal sets, OOD, and Safe Mode	16
6.5	Explanation contract	17
7	Dynamic Action Window and Next Best Observation	17
7.1	Action Window derivation	17
7.2	Next Best Observation	17
8	Queue Twin, Clinical Slack, and Scheduling Policy	18
8.1	Queue Twin	18
8.2	Clinical Slack	18
8.3	Primary queue result	19
8.4	Response-definition sensitivity	19
8.5	Alert workload	20
9	Software and Product Architecture	20

9.1	Deployment units	20
9.2	Product interaction model	21
9.3	Audit mechanism	23
10	Evaluation Protocol	23
10.1	Predeclaration and research questions	23
10.2	Policies	24
10.3	Endpoints and reporting	24
11	Results and Validation	24
11.1	Model and subgroup results	24
11.2	Queue and operational results	24
11.3	NBO negative finding	25
11.4	External input plausibility	25
11.5	Engineering assurance	25
11.6	Reproducibility hashes	26
12	Independent Review and Corrective Engineering	26
12.1	First review: pre-product red team	26
12.2	Second review: release-candidate panel	27
13	Governance, Privacy, and Human Accountability	28
14	Reproducibility and Deployment	28
14.1	Local developer workflow	28
14.2	Container contract	28
14.3	Vercel deployment architecture and live acceptance	29
15	Limitations and Threats to Validity	29
15.1	Construct validity	29
15.2	Internal validity	29
15.3	External validity	30
15.4	Human factors	30
15.5	Security and regulatory readiness	30
15.6	Clinical-safety boundary	30
16	Prospective Validation and Product Roadmap	30
17	Conclusion	31
A	Clinical Calculation Pseudocode	32

B API Surface	32
C Claim-Control Ledger	33
D Artifact and Evidence Map	33
E Reproduction Checklist	34

List of Figures

1	System architecture	6
2	Runtime decision pipeline	7
3	The differentiation claim is a recomposition of established ideas, not invention of the individual components.	11
4	ML development and validation pipeline	14
5	Selected-model recall and calibration by horizon. Stress ECE deterioration motivates explicit shift handling.	16
6	NBO gate outcome. The efficiency gain did not preserve the fixed safety boundary. . .	18
7	Site and overall Action Window miss rates under simulated 3x surge.	19
8	Post-verification response-definition sensitivity and paired confidence intervals.	20
9	Working Deadline Board after a deterioration event. The patient-level deadline and queue-level capacity conflict are shown together without collapsing them into one score.	22
10	Human authority is a visible product state, not a disclaimer.	22
11	Scientific evidence and operational constraints are available from the working product.	22
12	Degraded mode suppresses live estimates and actions, labels last-verified context, and directs the downtime workflow.	23
13	Independent review cycle and material corrective outcomes.	26

List of Tables

1	Release-candidate evidence summary. Every quantitative performance result is synthetic or simulated.	8
2	Official problem-statement coverage and implemented evidence.	9
3	Core safety invariants and failure behavior.	12
4	Selected boosted-hazard performance by horizon on synthetic cohorts.	15
5	Prototype deployment units and responsibilities.	21
6	Automated test inventory at TL-07 release candidate.	25
7	TL-06.5 final-review disposition.	27

Visual technical map. The three principal blueprints are deliberately presented on dedicated pages: the complete layered system architecture (Figure 1), the online runtime decision pipeline (Figure 2), and the offline ML development/validation pipeline (Figure 4).

1 Executive Technical Synopsis

TriageLoop is built around a distinction that is easy to miss: clinical urgency and operational feasibility are related, but they are not the same variable. A patient's need determines when an action should occur. Capacity determines whether the department is likely to meet that requirement. Queue optimization must never make the clinical requirement appear less urgent.

The system therefore answers six questions in sequence:

1. What has changed in this waiting patient?
2. Do deterministic age-aware safety rules require immediate action?
3. What is the calibrated trajectory of near-term escalation risk, and how uncertain is it?
4. By when should the next specified clinical action occur?
5. Given current resources and queue state, will that time requirement be met?
6. What does the clinician decide, and what evidence must be preserved?

The corresponding operating quantities are:

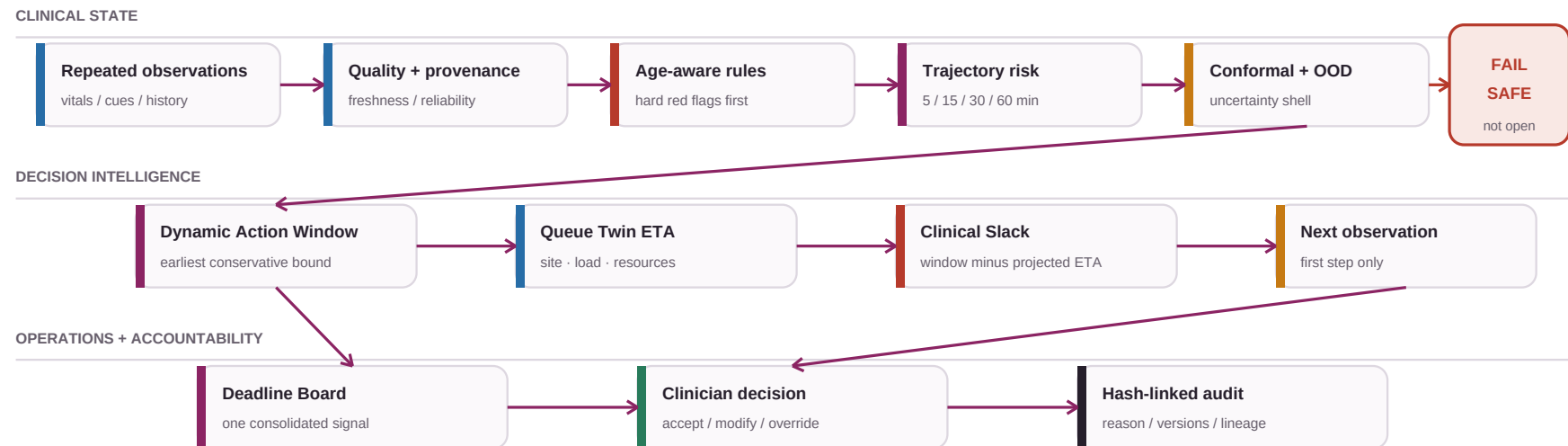
$$W_i(t) = \min\{W_{\text{rule}}, W_{\text{category}}, W_{\text{model}}, W_{\text{safe mode}}\}, \quad (1)$$

$$S_i(t) = W_i(t) - \hat{\eta}_i(t), \quad (2)$$

where $W_i(t)$ is the maximum recommended delay to the next action for patient i , $\hat{\eta}_i(t)$ is the Queue Twin's projected time to that action, and $S_i(t)$ is Clinical Slack. Negative Slack is not a diagnosis; it is a visible capacity conflict.

TriageLoop end-to-end safety architecture

Patient need becomes time; time is tested against capacity; a clinician closes the loop.



Safety invariants: rules cannot be suppressed / no autonomous downgrade / feasibility never changes need / audit failure blocks state change

Figure 1: SYSTEM ARCHITECTURE. End-to-end layers, feedback paths and safety invariants. Clinical need is calculated before operational feasibility.

Runtime clinical-decision pipeline

Every update recomputes need first, then tests queue feasibility; the clinician owns the action.

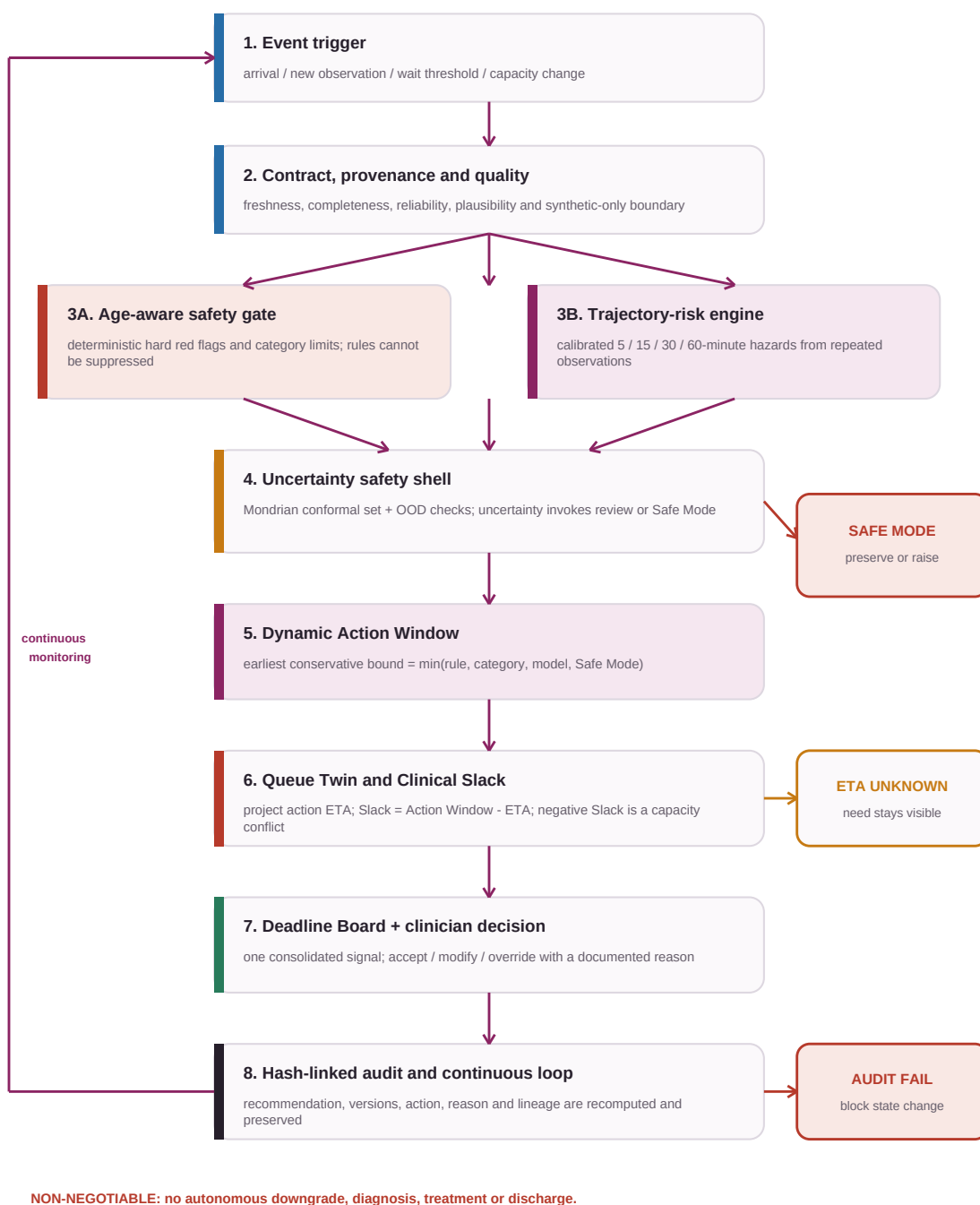


Figure 2: RUNTIME PIPELINE. Event-to-action flow for every arrival, new observation, elapsed-wait trigger or capacity change.

Central contribution. TriageLoop does not claim “AI triage” as novel. Its defensible contribution is a coherent waiting-room control loop in which repeated-observation risk constrains a conservative deadline, uncertainty invokes a specific safe response, a Queue Twin tests feasibility without rewriting clinical need, and a clinician closes an auditable loop.

1.1 Evidence snapshot

Table 1: Release-candidate evidence summary. Every quantitative performance result is synthetic or simulated.

Evidence unit	Result	Interpretation boundary
Trajectory model	Test recall 0.902-0.975; stress recall 0.891-0.982 across four horizons	Internal consistency on generated cohorts; not clinical accuracy
Periodic comparator	17.7% to 14.1% Action Window misses; 20.5% relative reduction	1,200 paired synthetic policy shifts at 3x load
Response sensitivity	12.8% / 13.0% / 24.1% relative reductions at 10 / 20 / 30 minutes	Post-verification robustness analysis; only 30 minutes exceeds 20%
NBO hypothesis	87.5% fewer observations, but -7.1 percentage-point critical recall	Failed safety gate; feature demoted to first-step guidance
External plausibility	63,163 open MIMIC-IV demo measurements across six mapped vital fields	Input scale only; adult hospital/ICU demo, not ED transfer
Engineering assurance	73/73 Python tests; type-check and optimized web build pass	Prototype verification; not certification

2 Problem Definition and Official-Brief Coverage

2.1 The waiting-room failure mode

The official PatientTriage.ai brief asks for an assistant that helps staff prioritize and route arriving patients without replacing clinical judgment, while handling age variation, incomplete information, asymmetric under-triage cost, explainability under time pressure, surge behavior, reassessment of waiting patients, override, audit, protection of patient data, and cross-hospital variability [1]. The minimum demonstration requires at least 15-20 simulated records, an ambiguous presentation, a pediatric or geriatric case, a zero-history patient, a 3x surge, explicit uncertainty, and a clinician override with its log.

A static arrival score does not fully address that specification. After intake, three coupled processes continue:

- the patient’s observed state evolves;
- evidence quality changes as new measurements appear or become stale;
- capacity and queue position change as arrivals and service completions occur.

TriageLoop treats the waiting room as a closed-loop safety-control problem. It does not claim to solve

ED crowding or create resources. It makes the mismatch between patient-specific need and available capacity explicit soon enough for a clinician or charge nurse to act.

2.2 Requirement-to-mechanism traceability

Table 2: Official problem-statement coverage and implemented evidence.

Official complexity / expectation	TriageLoop mechanism	Implemented evidence
Ambiguous and overlapping symptoms	Multimodal structured state, conformal prediction sets, Safe Mode, first-step NBO	Curated ambiguity scenarios; uncertainty visible in board and inspector
Age-specific thresholds	Pediatric/adult/geriatric stratification; deterministic age-aware safety rules	Rule catalogue, subgroup evaluation, pediatric and geriatric fixtures
Inconsistent intake data and zero history	Strict provenance, completeness/reliability score, freshness and plausibility checks	Zero-history contract test; missing/stale/implausible Safe Mode scenarios
Explainable decisions within seconds	Action-first explanation: what changed, why it matters, reliability, next action	Persistent patient inspector; corrected observed-positive model factors
Asymmetric under-/over-triage cost	5:1, 10:1 and 20:1 threshold sweeps; 10:1 selected subject to recall gate	Low operating thresholds, recall-first selection, no autonomous downgrade
Different hospitals and staffing	Configured community, regional, and urban/trauma site profiles	100/300/550 visits-day assumptions; baseline and 3x simulations
Reviewable and overridable recommendation	Accept, modify, override; reason required for material divergence	Browser-verified override and audit receipt
Integration maturity varies	Versioned JSON contracts, manual synthetic intake, FHIR-like boundary, SSE with fallbacks	Strict intake endpoint, schema tests, local API and proxy layer
Waiting-patient monitoring	Repeat observations trigger recomputation; absolute bounds age with waiting time	Deterioration event promotes P-0009 and produces -8 minutes Slack
Safe-wait threshold breach	Category bound and Action Window expire independently of new model output	Queue aging and capacity-conflict scenario tests
Patient-data protection	Synthetic-only contract, pseudonymous IDs, minimization, narrow CORS and headers	Direct name rejected; secrets/raw data/database ignored; India governance baseline
Adoption and alert fatigue	One consolidated current action per patient; measured alert denominators; degraded mode	0.86-1.12 signals/waiting-patient-hour during surge; Community warning retained
15-20 simulated patients	Twelve-patient live board plus 28 curated fixtures and 10,000 generated encounters	Automated count/coverage tests and deterministic demo reset

Official complexity / expectation	TriageLoop mechanism	Implemented evidence
3x surge	Queue Twin with site/load configuration and scenario switch	1,800 original and 1,200 stronger-comparator policy shifts
Explicit confidence	Horizon risks, conformal set, uncertainty state, OOD, abstention and quality	Recommendation contract and live inspector
Clinician override and log	Reasoned state transition plus hash-linked event	Five-event live chain; tamper detection test

3 Research Lineage and Differentiation

3.1 Clinical and machine-learning foundations

Repeated measurements can carry information that a single arrival observation cannot. A prospective ED study reported that repeated vital-sign measurements better identified deterioration than admission values alone in its infection/sepsis cohort [2]. A later systematic review found potential added value in vital-sign trends but also low-certainty and sparse evidence, a boundary that argues against overstating transferability [3]. These findings motivate longitudinal inputs; they do not validate TriageLoop.

Time-varying deterioration modeling and prognosis-informed prioritization provide adjacent methodological precedent [4, 5]. TriageLoop adopts discrete-time horizon risk because the output can be calibrated, inspected, and translated into bounded action times. It does not use an LLM in the quantitative clinical path.

Uncertainty is represented explicitly because a score without a reliability state would conflict with the brief. Conformal prediction offers a distribution-free finite-sample framework under exchangeability assumptions [6]; a 2025 ED disposition study demonstrated the operational “I don’t know” pattern with conformal prediction [7]. TriageLoop uses class-conditional prediction sets so rare critical cases are not hidden by aggregate coverage.

3.2 Cross-domain transfer

Three mechanisms are deliberately borrowed from outside conventional triage dashboards:

1. **Clinical deadline from prognostics.** Remaining Useful Life estimates time remaining before an equipment state becomes unacceptable [8]. TriageLoop translates the intuition, not the literal estimator, into a conservative next-action window. It avoids phrases such as “safe until” because clinical time is not a known mechanical failure time.
2. **Clinical Slack from real-time scheduling.** Deadline and laxity scheduling asks how much scheduling freedom remains before a deadline is missed [9]. TriageLoop computes Slack as patient-specific time requirement minus projected action time, while retaining deterministic category and rule precedence.
3. **Next Best Observation from active feature acquisition.** Active acquisition asks which costly feature would most improve a case-specific decision [10]. The prototype restricts candidate actions to low-burden observations and questions; it does not autonomously order treatment, imaging, or laboratory tests.

3.3 Prior art and novelty boundary

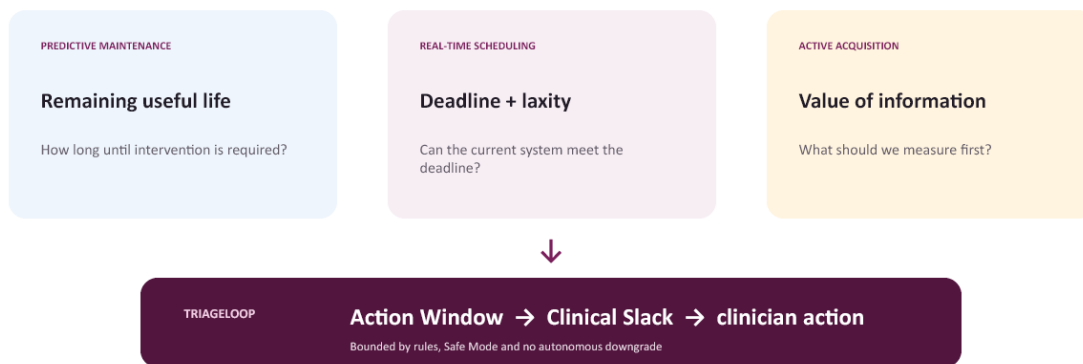
Dynamic clinical queues, continuous reassessment, and AI-supported triage all have precedent. The closest identified 2026 India-focused OPD work combines LLM-based severity prediction, urgency-drift detection, longitudinal memory, and adaptive queue optimization in simulation [12]. TriageLoop therefore does not claim “world first,” patent novelty, or ownership of dynamic queues.

The narrower distinction is the system-level composition shown in Figure 1: an ED waiting-room workflow; deterministic rules before ML; no LLM in the numerical path; calibrated multi-horizon trajectory risk; conformal/OOD Safe Mode; a conservative Action Window; explicit capacity feasibility through Slack; NBO constrained by a failed safety gate; no autonomous downgrade; and clinician-linked audit. Novelty is claimed as a design recomposition and working prototype, not as clinical validation.

WHY THIS IS DIFFERENT

The wow factor is the recomposition—not an ‘AI triage’ claim.

Three proven ideas from other domains become one safety-oriented ED waiting-room control loop.



Synthetic decision-support prototype — not validated for clinical use

08

Figure 3: The differentiation claim is a recomposition of established ideas, not invention of the individual components.

4 Safety Contract and System Invariants

The clinical safety contract was locked before implementation. Its purpose is to define what the prototype may never do, independent of model performance.

Table 3: Core safety invariants and failure behavior.

Invariant	Normal behavior	Failure behavior
Rules precede model	Hard red flags immediately constrain action	Model output cannot suppress or lengthen the rule requirement
No autonomous downgrade	Clinician-assigned state is preserved unless a clinician changes it	Model may recommend earlier review or promotion only
Uncertainty escalates	Ambiguous/OOD/poor-quality state is named and prompts reassessment	Safe Mode uses a conservative bound and specific reason
Need is separate from capacity	Action Window remains a clinical requirement	Queue failure removes ETA/Slack confidence, not the patient deadline
Human authority is explicit	Accept, modify, and override are available	Modify/override requires reason; persistence failure blocks the digital action
Audit is verified, not asserted	Every event stores previous hash and event hash	Integrity endpoint recomputes all links and reports first broken event
Degraded mode is honest	Live state and controls are available only with service connection	UI labels last-verified context, suppresses estimates, disables actions, and directs downtime workflow

When multiple triggers occur, the system does not emit competing actions. It selects the most conservative action and preserves all contributing reasons. A compound fixture combines a hard red flag, Safe Mode, and an exceeded wait threshold to verify this behavior.

The prototype’s deterministic thresholds are conservative demonstration rules, not a deployable clinical protocol. Pediatric separation is motivated by WHO’s separate child and adult/adolescent facility triage tools [11]; geriatric handling adds frailty, comorbidity, and atypical-presentation context but still requires local clinical governance.

5 Data Strategy and Synthetic Longitudinal Cohort

5.1 Why synthetic data is primary

The competition does not require proprietary hospital data and explicitly permits realistic simplified or simulated records [1]. Synthetic generation was chosen to guarantee pediatric, geriatric, ambiguous, missing-data, zero-history, and surge cases while avoiding identifiable patient information and access delays. This choice enables reproducibility but creates the paper’s largest validity threat: the model, generator, and Queue Twin originate in one project.

The paper therefore distinguishes three evidence classes:

- **curated scenario evidence** verifies required and adversarial behavior;
- **synthetic statistical evidence** checks internal model and queue hypotheses under declared assumptions;
- **external plausibility evidence** checks only whether input scales are grossly disconnected from an open reference.

5.2 Population and split

The locked generator uses seed 20260821 to create 10,000 longitudinal encounters and 30,020 time-indexed snapshots. The final snapshot allocation is 17,114 train, 5,685 validation, 5,708 untouched test, and 1,513 deliberate stress snapshots. Splits are patient-level. Validation is internally separated so probability calibration and conformal/threshold selection do not reuse the same examples.

History availability is approximately balanced: 4,921 encounters have some prior history and 5,079 have none. The generated population spans pediatric, adult, and geriatric bands; the stress cohort changes age mix, missingness, prevalence, arrival load, and measurement noise.

5.3 Observation contract and temporal leakage controls

Each patient state contains a pseudonymous identifier, age, sex at birth, complaint, symptoms, observed cues, history availability, clinician state, and an ordered observation series. Every observation records time, source, vital values, completeness, reliability, and synthetic provenance. Future observations and hidden deterioration labels are excluded from features at prediction time.

Forty-four standardized/imputed features include current values, deltas, missing indicators, age bands, history state, clinician level, elapsed wait, data quality, and bounded text-cue flags. The feature “number of observations” was removed after audit because it could encode workflow intensity rather than physiology. This is an example of process-derived leakage: a model can appear predictive by learning how often staff already chose to measure a patient.

Trajectory-risk development and safety pipeline

All development evidence is synthetic unless explicitly labelled as external input plausibility.

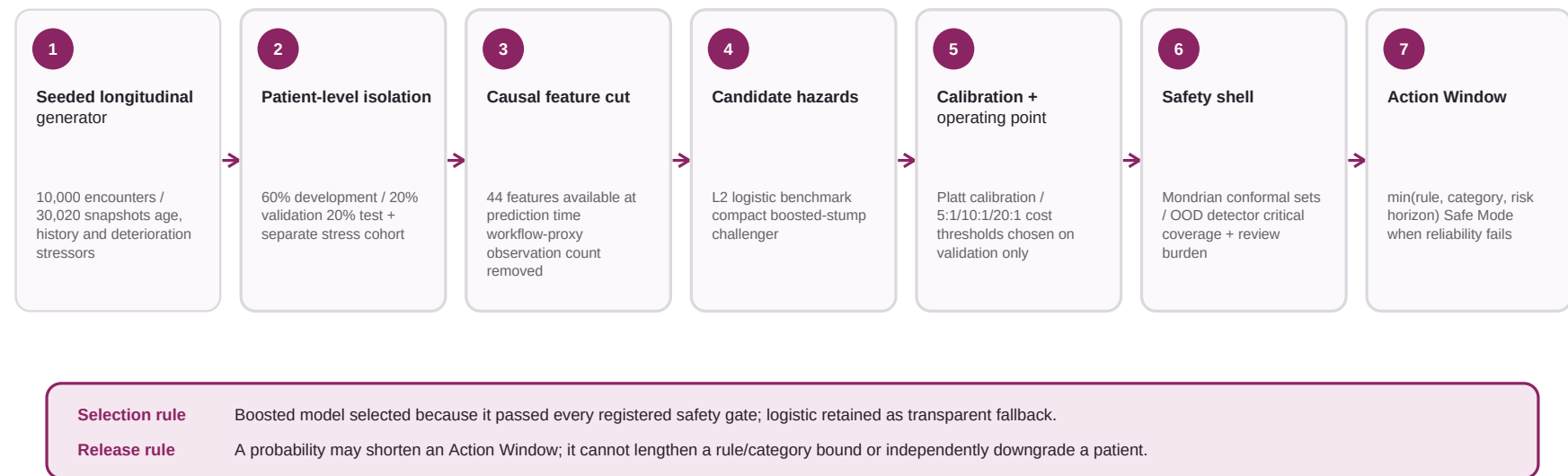


Figure 4: ML DEVELOPMENT AND VALIDATION PIPELINE. Data generation, patient-level isolation, leakage control, candidate comparison, calibration, uncertainty gates and release rule.

5.4 Curated cases

Twenty-eight deterministic fixtures cover ambiguous presentations, pediatric and geriatric states, no-history intake, missing and stale data, worsening vital signs, hard-rule/model conflict, compound triggers, multiple critical patients, NBO unavailability, and override/audit behavior. Twelve of these appear on the live board to keep the jury demonstration readable while the full set remains in automated tests.

6 Trajectory-Risk and Uncertainty Methodology

6.1 Prediction target

For each snapshot $x_{i,t}$, the model estimates simulated need for urgent clinical escalation within horizons $h \in \{5, 15, 30, 60\}$ minutes:

$$p_{i,t,h} = P(T_i \leq t + h \mid T_i > t, x_{i,t}). \quad (3)$$

The generated composite can represent a resuscitation-level vital abnormality, urgent senior review, monitored-care requirement, or treatment-space escalation. It is not time to death, a diagnosis, or a promise of safety until a particular minute. Horizon probabilities are monotonized after calibration so longer horizons do not report less cumulative risk than shorter ones.

6.2 Candidates and model selection

The primary benchmark is an L2-regularized discrete-time logistic hazard model. The challenger is a compact gradient-boosted ensemble of shallow logistic decision stumps. Both use the same feature transformer and Platt calibration. AUROC alone is prohibited as a selection criterion. The challenger may win only through a meaningful safety/calibration advantage without unacceptable subgroup or explanation burden.

The boosted model was selected by a registered gate override. The logistic benchmark had slightly better mean test Brier score (0.0476 versus 0.0498), but failed short-horizon/stress gates after removal of the workflow-proxy feature. The boosted model passed every registered horizon-level recall, in-distribution ECE, and critical-class conformal-coverage gate. Its mean stress recall was 0.941 versus 0.862 for logistic.

Table 4: Selected boosted-hazard performance by horizon on synthetic cohorts.

Horizon	Test recall	Stress recall	Test ECE	Test critical coverage	Stress critical coverage	Test positive rate
5 min	0.914	0.891	0.003	0.973	0.977	0.080
15 min	0.902	0.923	0.008	0.902	0.923	0.267
30 min	0.975	0.982	0.016	0.975	0.982	0.545
60 min	0.949	0.968	0.026	0.949	0.968	0.523

Selected-model performance across time horizons

Synthetic untouched test and deliberately shifted stress cohorts; recall is prioritised over average discrimination.

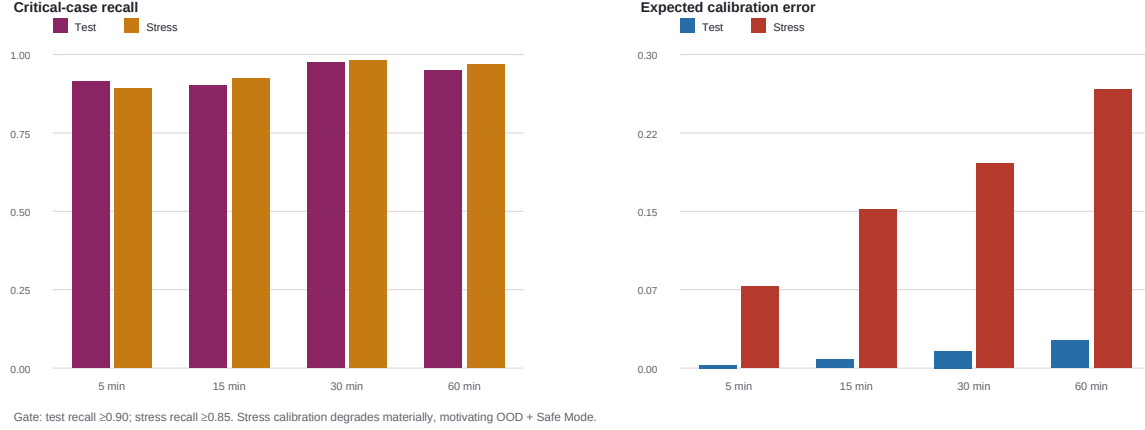


Figure 5: Selected-model recall and calibration by horizon. Stress ECE deterioration motivates explicit shift handling.

6.3 Asymmetric decision cost

Thresholds are selected on validation data at cost ratios $C_{FN} : C_{FP} \in \{5 : 1, 10 : 1, 20 : 1\}$ with a minimum validation recall constraint. The chosen prototype setting is 10:1:

$$\tau_h^* = \arg \min_{\tau} (C_{FN} FN_h(\tau) + C_{FP} FP_h(\tau)) \quad \text{subject to} \quad \text{Recall}_h(\tau) \geq 0.90. \quad (4)$$

The resulting thresholds are 0.4100, 0.0393, 0.0600, and 0.1225 at 5/15/30/60 minutes. The low 15- and 30-minute thresholds make the intended escalation bias visible. They also increase review workload, so the product consolidates multiple model/rule states into one current patient action and reports alert denominators rather than calling the burden manageable.

6.4 Conformal sets, OOD, and Safe Mode

The prototype uses 90% nominal class-conditional (Mondrian) conformal prediction sets. For class y and nonconformity score $s(x, y)$, the prediction set is

$$\Gamma_{\alpha}(x) = \{y : s(x, y) \leq q_{y, 1-\alpha}\}, \quad (5)$$

where $q_{y, 1-\alpha}$ is calculated from the isolated conformal calibration subset. A singleton set communicates a bounded prediction; a set containing both labels is ambiguous; an empty or materially ambiguous near-term set triggers review. Coverage is evaluated separately for the critical class because overall coverage could conceal rare-class failure.

A feature-space OOD detector flags 1.1% of test snapshots and 18.2% of stress snapshots. Missing, stale, implausible, low-reliability, OOD, or materially ambiguous near-term states activate Safe Mode. Safe Mode does not invent a more precise probability. It recommends conservative review, states why the numerical path is unreliable, and prevents automated downgrade.

6.5 Explanation contract

The nurse-facing explanation answers four questions: what action is recommended, by when, what changed, and why the system trusts or distrusts its estimate. Local stump contributions are translated only when they are positive urgency drivers grounded in observed values. A TL-06.5 review found that an absent binary feature could receive a mathematically valid negative centered contribution and be verbalized as if the finding were present. The numerical score was unchanged, but the sentence was clinically backward. The renderer was corrected to admit only observed positive drivers, and a regression test now checks the deteriorating P-0009 case.

7 Dynamic Action Window and Next Best Observation

7.1 Action Window derivation

The Action Window is a conservative interval by which the next specified action is recommended. For patient i at time t ,

$$W_i(t) = \min [W_{\text{hard},i}(t), \max(0, W_{\text{cat},i} - e_i(t)), \inf\{h : p_{i,t,h} \geq \tau_h\}, W_{\text{safe},i}(t)], \quad (6)$$

where $e_i(t)$ is elapsed wait, the category term is an upper bound, the model term is the earliest calibrated threshold crossing, and Safe Mode contributes a five-minute maximum in the prototype. Missing terms are treated as $+\infty$. A hard red flag yields immediate review. The interface says “recommended within”; it never says “safe until.”

This design converts a probability vector into a workflow constraint without pretending that the model predicts an exact deterioration moment. It also makes waiting-time aging explicit: even without a new observation, a category-bound window shrinks as elapsed time increases.

7.2 Next Best Observation

When uncertainty is decision-relevant and no hard rule already determines the action, the system ranks a clinician-approved catalogue containing repeat SpO2, respiratory rate, heart rate, blood pressure, temperature, mental status/GCS, pain, and symptom-onset clarification. A conceptual value function is

$$o^* = \arg \max_{o \in \mathcal{O}} \frac{\mathbb{E}[\Delta D \mid o]}{t_o + \lambda b_o}, \quad (7)$$

where ΔD is estimated decision impact, t_o is typical acquisition time, and b_o is burden. The implementation combines model feature importance, proximity to the decision threshold, and quality defects. Candidate observations are suggestions only; no treatment, laboratory, or imaging order is generated.

The predeclared gate required at least 25% fewer observations with no more than two percentage points of operational critical-recall loss versus an eight-observation bundle. The one-observation policy failed decisively on 4,317 eligible snapshots: recall fell from 0.959 to 0.888. A four-observation adaptive fallback passed the test split but failed independent stress confirmation. Selecting a five-observation bundle after viewing stress would be post-hoc tuning, so it was rejected.

Next Best Observation: efficiency hypothesis rejected

Counterfactual on 4,317 eligible synthetic snapshots; gate fixed before evaluation.



Figure 6: NBO gate outcome. The efficiency gain did not preserve the fixed safety boundary.

Release consequence. The product may show NBO as the first measurement suggestion. If the measurement is unavailable, uncertainty persists, or clinical concern remains, the required behavior is full reassessment or escalation. No workload-reduction or recall-preservation claim is permitted.

8 Queue Twin, Clinical Slack, and Scheduling Policy

8.1 Queue Twin

The Queue Twin is a seeded discrete-event simulator and live projection layer. It estimates time to the next clinical action using current queue state, site profile, load, clinician capacity, reassessment staff, monitored spaces, service-time distributions, and priority policy. It is not a staffing optimizer and is not calibrated to a real hospital.

The original registered matrix contains three policies, three sites, two loads, and 100 paired replications per cell: 1,800 policy shifts. TL-05 added a stronger two-policy matrix: fixed 15-minute periodic re-triage versus TriageLoop across the same sites, loads, and replications, totaling 1,200 policy shifts.

8.2 Clinical Slack

For patient i ,

$$S_i(t) = W_i(t) - \hat{\eta}_i(t). \quad (8)$$

$S_i > 0$ means projected capacity can meet the current window under the simulator’s assumptions. $S_i = 0$ means the deadline is due. $S_i < 0$ means the current queue is projected to act after the requirement. The UI labels the latter “Capacity conflict” and asks for operational escalation; it does not relax the patient deadline.

The production policy is no-demotion and lexicographic. Hard rules determine immediate action. The dynamic category follows, allowing strong repeat-observation risk to promote but not autonomously demote. Absolute Action Window, calibrated risk, uncertainty, and arrival time resolve later ties. This policy was selected after pure earliest-deadline scheduling grouped excessive false-positive 30-minute windows and failed the registered gate.

8.3 Primary queue result

Under simulated 3x load, TriageLoop reduced Action Window misses from 17.7% to 14.1% versus fixed 15-minute periodic re-triage, a 20.5% relative reduction (95% paired-bootstrap CI 17.4%-23.8%). It also reduced negative-Slack minutes by 14.0%, signal-to-action time by 15.7%, and low-acuity P90 wait by 14.4% in the same overall comparison.

Action Window misses under simulated 3x surge

Paired comparison against fixed 15-minute periodic re-triage; 1,200 policy shifts across three site profiles.

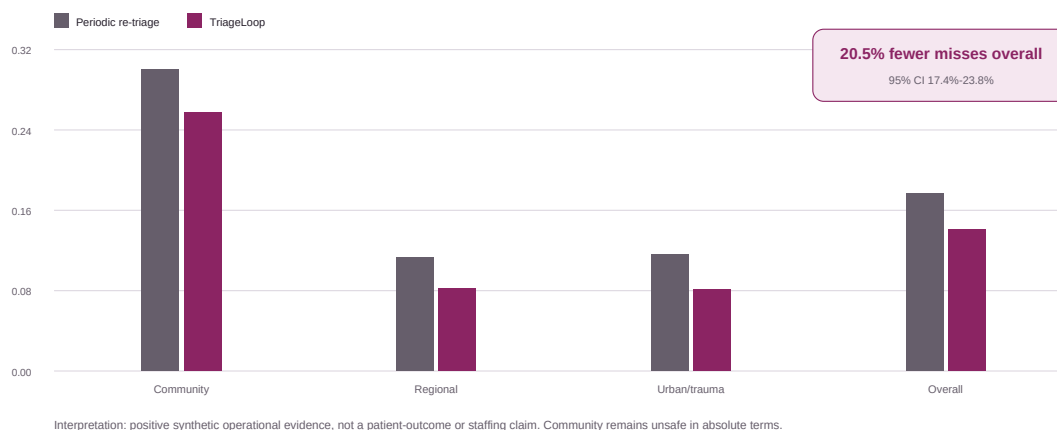


Figure 7: Site and overall Action Window miss rates under simulated 3x surge.

The result is not uniform proof of safety. Under baseline capacity all compared policies have zero critical misses, so TriageLoop has no measurable miss-rate advantage. Community surge remains unsafe in absolute terms despite improvement. In the earlier static comparison its miss rate remains 25.7% and critical recall 0.743. Software cannot create staff, beds, or monitored spaces; the product must keep negative Slack visible.

8.4 Response-definition sensitivity

The composite deterioration target includes immediate red flags and broader urgent review/monitoring/treatment escalation. The simulation therefore reports sensitivity to a 10-, 20-, and 30-minute response allowance. Against the stronger periodic comparator, relative miss-rate reductions were 12.8%, 13.0%, and 24.1%, with positive paired intervals at each definition. Only the 30-minute definition exceeds 20%. This analysis was added after TL-05 in response to independent review and is not a retroactively registered gate.

Response-definition sensitivity

Post-verification robustness analysis against periodic re-triage; not a retroactively registered gate.

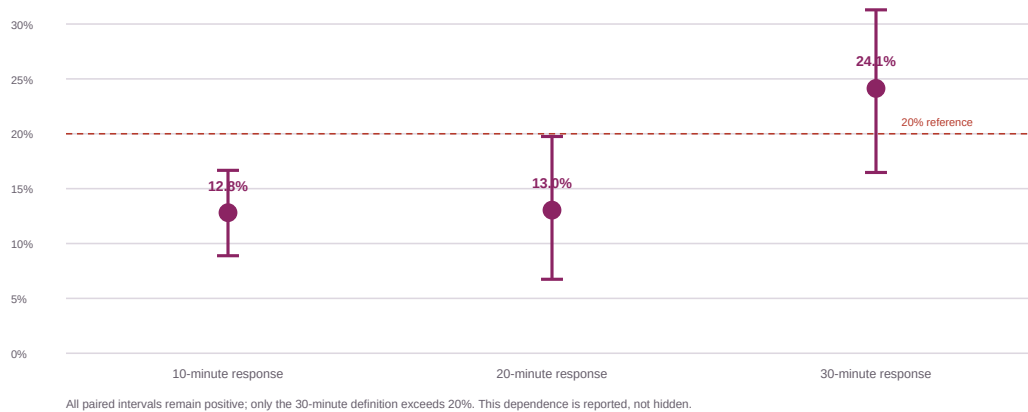


Figure 8: Post-verification response-definition sensitivity and paired confidence intervals.

8.5 Alert workload

Signals are consolidated at patient level within each eight-hour shift. During surge, TriageLoop produces 0.86-1.12 consolidated alerts per waiting-patient-hour and 2.20-5.56 alerts per configured reassessment-nurse-hour across site profiles. Community surge is the warning case at 5.56 signals per nurse-hour under the one-nurse assumption. No “safe workload” threshold is claimed; real nurse time-and-motion and alert-acknowledgement studies remain external work.

9 Software and Product Architecture

9.1 Deployment units

The system is separated into a nurse-facing Next.js application, a FastAPI decision/simulation service, versioned JSON contracts, and a SQLite prototype store. The frontend never recomputes clinical logic. It calls a same-origin server proxy, which forwards to the configured API. The API owns validation, recommendation, simulation state, clinician decisions, and audit integrity.

Table 5: Prototype deployment units and responsibilities.

Unit	Technology	Responsibility
Nurse interface	Next.js App Router / TypeScript	Deadline Board, patient inspector, Capacity, Audit, Evidence, About/resilience surfaces
Decision service	Python 3.12 / FastAPI	Contracts, rules, model inference, uncertainty, Action Window, manual synthetic intake, decisions
Operational layer	Seeded discrete-event simulation	Queue ETA, site/load scenarios, Slack, paired evaluation
Prototype store	SQLite	Deterministic demo state, decision state, append-only hash-linked audit events
Contracts	JSON Schema / Pydantic	Strict patient, recommendation, and queue snapshot boundaries
Container path	Docker Compose	API health-gated startup followed by standalone Next.js service

The API exposes health/configuration, demo state and reset, scenario changes, deterioration events, patient/recommendation retrieval, manual synthetic intake, decisions, audit, audit-integrity verification, evaluation artifacts, and a server-sent event stream. CORS permits only configured origins, state-changing methods are restricted, and request models reject unknown direct identifier fields.

9.2 Product interaction model

The primary surface is a clinical Deadline Board, not a generic analytics dashboard. Each row shows rank, pseudonymous patient, age band, complaint, current level, action, Action Window, ETA, Slack, uncertainty, and latest change. A persistent inspector follows a fixed order: recommendation; deadline/ETA/Slack; trajectory; reasons and uncertainty; first-step NBO; clinician decision.

Meaning never depends on color alone. Red is reserved for clinical urgency and capacity conflict; brand berry indicates selection and primary actions. The system targets WCAG 2.2 AA structure through one H1, skip link, labeled controls, table captions/headers, visible focus, reduced motion, and text equivalents. This is structural prototype evidence, not formal assistive-technology or nurse-usability validation.

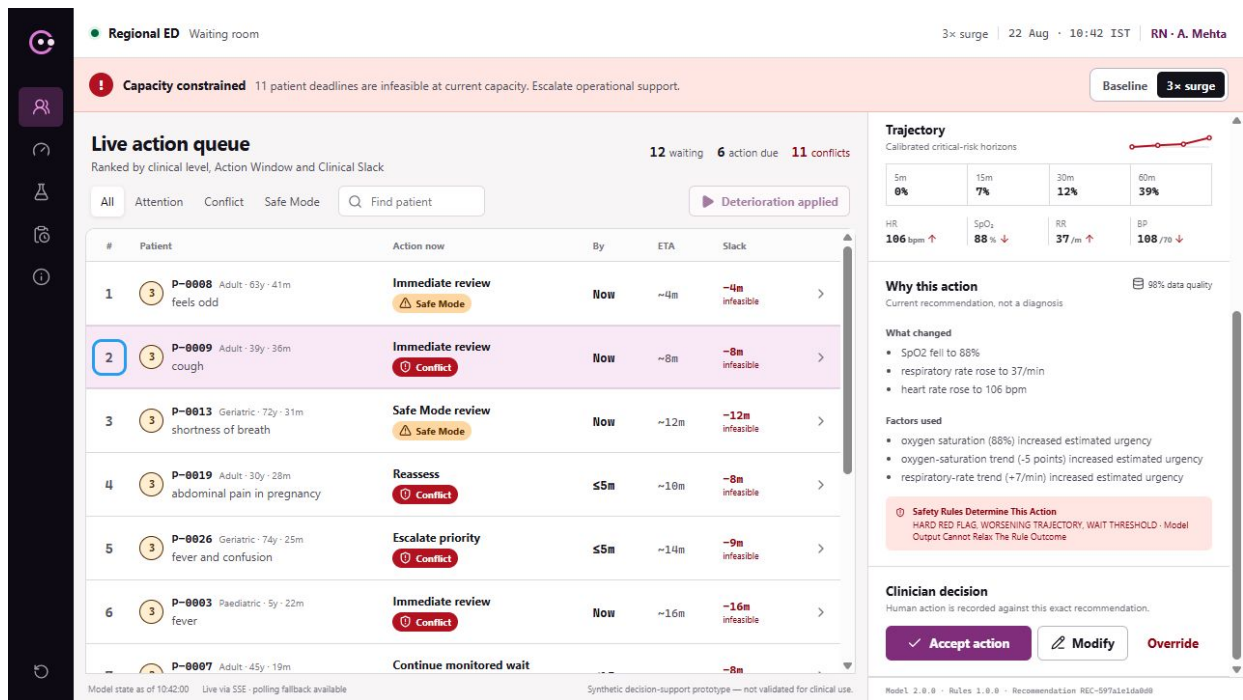
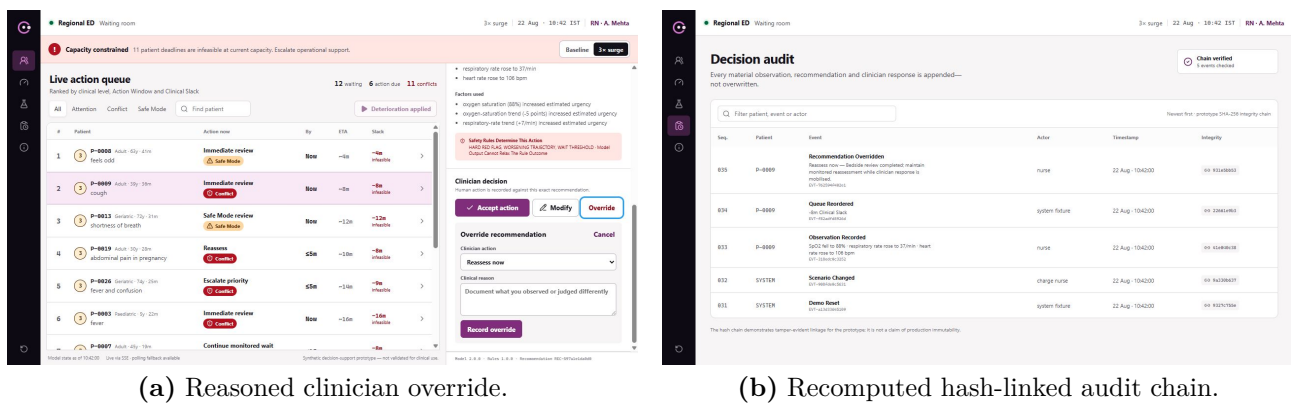


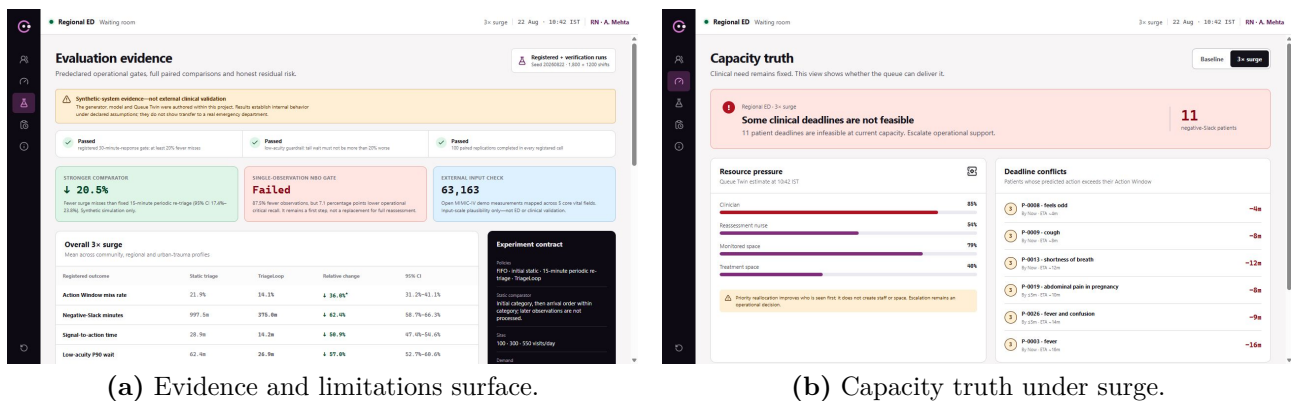
Figure 9: Working Deadline Board after a deterioration event. The patient-level deadline and queue-level capacity conflict are shown together without collapsing them into one score.



(a) Reasoned clinician override.

(b) Recomputed hash-linked audit chain.

Figure 10: Human authority is a visible product state, not a disclaimer.



(a) Evidence and limitations surface.

(b) Capacity truth under surge.

Figure 11: Scientific evidence and operational constraints are available from the working product.

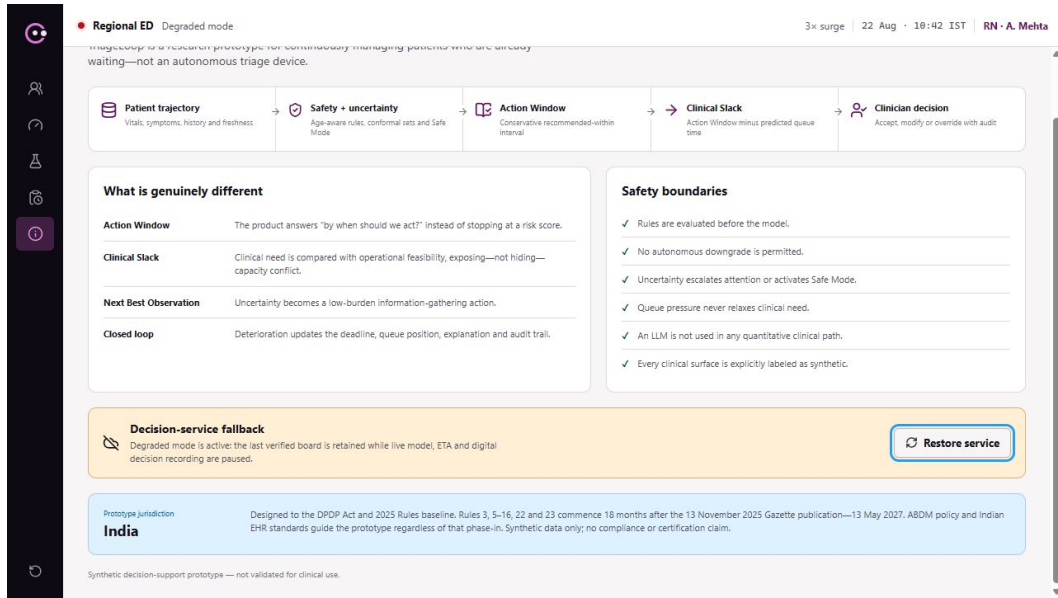


Figure 12: Degraded mode suppresses live estimates and actions, labels last-verified context, and directs the downtime workflow.

9.3 Audit mechanism

Each event contains a unique ID, optional patient ID, timestamp, actor, event type, structured payload, previous hash, and event hash. With canonical serialization $c(e_k)$,

$$H_k = \text{SHA256}((H_{k-1} \text{ or GENESIS}) \parallel c(e_k)). \quad (9)$$

Integrity is reported only after recomputing every event hash and previous-link value in sequence. A direct database-tampering test verifies that modification is detected. This is tamper-evident prototype linkage, not immutable storage, non-repudiation, or a production security control.

10 Evaluation Protocol

10.1 Predeclaration and research questions

The evaluation protocol was locked before model and queue implementation to reduce metric cherry-picking. It asked whether TriageLoop could reduce missed Action Windows under crowding, preserve lower-acuity tail waits, expose uncertainty with bounded workload, use NBO to reduce observation burden without material recall loss, and remain directionally stable across age, history, site, and shift conditions.

The protocol predeclared the following prototype gates:

- in-distribution critical recall at least 0.90 and stress recall at least 0.85;
- critical-class conformal coverage at least nominal target minus 0.03;
- ECE no greater than 0.08 at the selected operating horizon;
- at least 20% fewer 3x-surge Action Window misses versus initial static triage;
- low-acuity P90 wait no more than 20% worse than static triage unless explicitly explained;
- NBO at least 25% fewer observations with no material critical-recall loss;
- 100% Safe Mode activation for hard-failure fixtures and 100% required scenario pass.

The stronger periodic comparator was added after an independent critique identified that the initial static policy did not process later observations. Its directional interpretation is reported separately; the original 20% threshold was not silently moved to this later comparison.

10.2 Policies

P0 - FIFO

Arrival order except immutable immediate red flags.

P1 - Initial static triage

Generic five-level category and arrival order within category; no later observation processing.

P1b - Fixed periodic re-triage

Every 15 minutes, accumulated observations are processed through deterministic rules/category logic; no learned trajectory risk, conformal uncertainty, Action Windows, or Slack.

P2 - TriageLoop

Rules, Dynamic Action Window, uncertainty escalation, no-demotion dynamic category, Queue Twin, Slack, and bounded NBO.

All policies replay identical arrivals, patient trajectories, service times, and resource availability for paired comparison. Paired bootstrap intervals are computed across shift replications.

10.3 Endpoints and reporting

Primary endpoints are critical Action Window miss rate, unsafe under-triage, signal-to-action time, critical recall, and negative-Slack minutes. Secondary endpoints include waiting-time and rank effectiveness, wait tails by acuity, throughput, queue length, utilization, conformal coverage, calibration, abstention/review burden, alerts per waiting-patient-hour, alerts per reassessment-nurse-hour, observations requested, and subgroup gaps.

Every machine-readable result includes synthetic/simulation boundaries. Positive, neutral, and negative results are retained. The selected model artifact and evaluation files carry deterministic seeds and SHA-256 hashes.

11 Results and Validation

11.1 Model and subgroup results

The selected boosted model meets the registered synthetic horizon gates shown in Table 3. At 30 minutes, pediatric/adult/geriatric test recall is 0.963/0.969/0.991; no/partial/available-history recall is 0.975/0.978/0.971. The worst-to-best observed subgroup range is 0.028. These cohorts inherit generator assumptions and do not establish fairness or clinical generalizability.

Stress recall remains high while stress ECE rises to 0.078-0.266 across horizons. This divergence matters: high recall does not make an absolute probability calibrated under shift. The correct response is OOD/Safe Mode and future local recalibration, not a stronger confidence claim.

11.2 Queue and operational results

The original registered static comparison showed 36.0% fewer 30-minute composite Action Window misses overall under 3x surge (95% CI 31.2%-41.1%). Because the comparator never re-triages, this

number combines the value of looking again with the value of learned dynamic prioritization. It is retained for provenance but is not the lead claim.

The lead result is the 20.5% relative reduction against fixed 15-minute periodic re-triage. Directional benefits also appear for negative Slack, signal-to-action time, and low-acuity P90 wait. Waiting-time and rank effectiveness do not improve uniformly in every cell; the strongest evidence is deadline adherence and lateness under the declared simulation.

11.3 NBO negative finding

The single-observation NBO policy reduces acquisition count and assumed acquisition time substantially, but violates the fixed recall boundary. The test-selected four-observation fallback fails stress confirmation. This result is a falsified feature hypothesis, not a system failure: the evaluation did what it was designed to do and converted an appealing efficiency feature into a bounded, safer product role.

11.4 External input plausibility

The open MIMIC-IV Clinical Database Demo v2.2 provides a 100-patient deidentified adult hospital/ICU subset [13]. The evaluation maps 63,163 measurements across heart rate, respiratory rate, SpO2, systolic and diastolic pressure, and temperature. All five core fields are present; at least 95% of each field lies within the prototype's hard plausibility ranges; each external median lies inside the synthetic p01-p99 interval.

This check rules out a gross unit/scale mismatch. It does not evaluate emergency-department transfer, pediatric behavior, labels, model calibration, deterioration prediction, Action Windows, queue performance, or patient outcomes. Full MIMIC-IV-ED is credentialed and was not used [14].

11.5 Engineering assurance

The final release-candidate suite contains 73 Python tests:

Table 6: Automated test inventory at TL-07 release candidate.

Test area	Files	Tests	Examples
Safety	3	13	Rule precedence, Safe Mode, explanation regression, observation catalogue
Evaluation	5	21	TL-02 gates, queue gates, periodic comparator, NBO, external plausibility
Product/API	2	14	Reset, deterioration, decisions, audit integrity, security boundaries
Unit	3	12	Generator, features, model mechanics
Simulation	1	6	Queue policy, capacity and scheduling behavior
Contract	1	4	Strict schemas and invalid input
Curated scenarios	1	3	Count/coverage, ambiguity, age and zero-history cases
Total	16	73	All passing

TypeScript type-check and the optimized Next.js production build pass. The isolated production rehearsal on ports 8100/3200 verified reset, deterioration, -8 minutes Slack, NBO boundary, 3x surge, reasoned override, audit integrity, evidence, and capacity without browser-console or page errors. Deliberately stopping the API produced the degraded state in Figure 12; explicit retry restored the live board after service restart.

Security-boundary tests reject a direct patient-name field, a 501-character override reason, unsupported DELETE reset, and malicious-origin CORS access. Frontend responses include content-type,

frame, referrer, and permissions-policy headers. These are local-prototype controls; production use requires authentication, authorization, TLS, secrets management, monitored storage, and hospital-specific threat modeling.

11.6 Reproducibility hashes

Repeated TL-05 executions reproduced evaluation artifacts byte-for-byte. Key hashes are:

- selected model: b37ab29077acd37865f3a9de89c070f37bbb549f8bc13f40d819146161a51990;
- periodic metrics: 6b4bed629305c5e447c46ebfeede5725870e88b614ebb0d882f6f3ac8fdd875e;
- NBO metrics: 2c5fc3840a988c5bffe1a79f100bbf6118dd19cc65771229abc34330da7b36d;
- external plausibility: 9e92598d67fd9f2f4254b3ac468a2399e56c6cacfd0ea90d26ab7836ce718c7.

12 Independent Review and Corrective Engineering

Two external Claude reviews were conducted at deliberately different maturity points. The first five-perspective panel reviewed the scientific and architectural package before the product was complete. The second six-perspective panel reviewed the working code, evidence, rendered proposal/deck, and release package before TL-07. Neither review was treated as an instruction source. Every finding was reproduced or disproved against code, machine-readable artifacts, the running product, and authoritative sources before implementation.

Independent review as an engineering control

Reviewer findings were verified against artifacts and code before adoption; critique was not treated as authority.

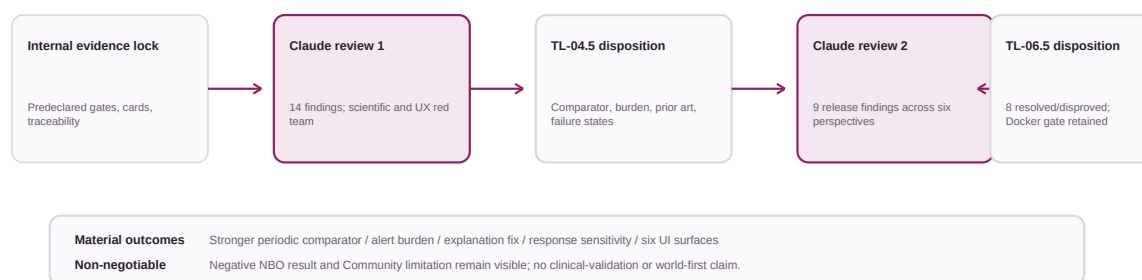


Figure 13: Independent review cycle and material corrective outcomes.

12.1 First review: pre-product red team

The first panel scored the concept as scientifically promising but fragile. Its central criticisms were circular synthetic evidence, framing drift toward queue optimization, an unfairly weak static comparator, unresolved nurse burden, close India-focused prior art, hidden sensitivity to the response definition, undefined compound-trigger behavior, and insufficient visible product evidence.

Several “not built” findings became obsolete after TL-04, but the scientific findings remained valid. The corrective actions were:

- reframe the story around a changing waiting patient before capacity;
- add the 15-minute periodic re-triage comparator;
- measure consolidated signals per waiting-patient-hour and reassessment-nurse-hour;
- cite the close India OPD prior art and narrow novelty claims;
- show Community separately and retain the absolute residual danger;
- make the synthetic evidence boundary unavoidable in the product and README;
- define compound-trigger precedence and degraded-mode behavior;
- run the NBO counterfactual rather than describing it as a benefit.

12.2 Second review: release-candidate panel

The final review issued a conditional go with nine findings. Eight were resolved or disproved:

Table 7: TL-06.5 final-review disposition.

ID	Verdict	Verified outcome
F-01	Valid; fixed	Corrected explanation attribution; regression test requires observed oxygen/trend factors and excludes absent confusion wording.
F-02	Not reproduced; strengthened	Alert evidence already existed; both denominators were surfaced consistently with Community as warning.
F-03	Valid; fixed	Added periodic-comparator 10/20/30-minute sensitivity: 12.8%/13.0%/24.1%.
F-04	Valid; resolved at release	Docker Desktop 4.87.0, Engine 29.7.2, Compose 5.4.0 and WSL 2.7.12 were installed. The built stack passed health, full browser, restart, persistence and reset checks.
F-05	Valid; fixed	Captured six working-product surfaces covering board, override, audit, evidence, capacity, and degraded mode.
F-06	Not reproduced	PMID 40318497 resolves to the intended 2025 uncertainty-aware ED triage paper.
F-07	Valid clarity issue; fixed	Distinguished the 1,800-shift TL-03 matrix from the 1,200-shift TL-05 matrix.
F-08	Valid clarity issue; fixed	Limited “zero findings” to the proposal-document audit; formal product accessibility/usability remains external.
F-09	Maintenance risk; fixed for release	Rechecked every source-register URL and replaced brittle links where needed.

The review process changed implementation and claims, not just prose. It found a real explanation bug, induced a new sensitivity experiment, strengthened comparator fairness, and expanded visible product evidence. It also demonstrates an important governance pattern: independent critique is most useful when paired with a written disposition that can say *accepted*, *not reproduced*, *superseded*, or *open*, with evidence for each.

13 Governance, Privacy, and Human Accountability

The assumed jurisdiction is India. The governance baseline references the Digital Personal Data Protection Act 2023 and its phased Rules, the Ayushman Bharat Digital Mission Health Data Management Policy, and Indian EHR Standards [15, 16, 17, 18]. The prototype is designed toward data minimization, purpose limitation, synthetic/pseudonymous identifiers, explicit audit events, configurable retention, and no secondary use. It does not claim legal compliance or certification.

No real patient data is required or permitted. Manual intake requires `provenance.synthetic=true`; unknown direct patient-name fields are rejected. The visible role label demonstrates workflow states, not production identity or authorization. A hospital deployment would require locally approved role definitions, identity federation, least privilege, encryption/key management, retention and deletion rules, consent/legal-basis analysis, incident response, monitored audit storage, and prospective clinical governance.

Human accountability is represented as state:

- the recommendation remains advisory;
- accept, modify, and override are distinguishable actions;
- modify/override requires a reason;
- the event records recommendation ID, model/rule versions, clinician action, reason, and audit linkage;
- no model or queue component may autonomously downgrade, discharge, treat, or diagnose.

14 Reproducibility and Deployment

14.1 Local developer workflow

From the repository root on Python 3.12+:

```
python -m venv .venv
.venv\Scripts\python -m pip install -e services/api
.venv\Scripts\python services/api/scripts/generate_data.py
.venv\Scripts\python services/api/scripts/train_models.py
.venv\Scripts\python services/api/scripts/run_queue_experiments.py
.venv\Scripts\python services/api/scripts/run_tl05_periodic_retriage.py
.venv\Scripts\python services/api/scripts/run_tl05_nbo.py
.venv\Scripts\python services/api/scripts/run_tl05_external_plausibility.py
Set-Location services/api
..\..\venv\Scripts\python -m unittest discover -s .././tests -p "test_*.py" -v
```

The API runs with Uvicorn on port 8000 and the Next.js application on port 3000. The deterministic jury path begins at `/board`; *Run deterioration event*, *3x surge*, and *Reset demo* exercise the principal states.

14.2 Container contract

The one-command path is:

```
docker compose up --build
```

Compose builds the Python API and standalone Next.js service. The web service starts only after the API health check at `/v1/health` passes. A named volume preserves prototype audit state; reset

always restores the deterministic demo. The acceptance gate was to reach /board from README-only instructions on a Docker-equipped machine, exercise deterioration/override/audit, restart Compose, and verify health plus expected persistence/reset behavior.

The closing deployment pass installed Docker Desktop 4.87.0 (Engine 29.7.2, Compose 5.4.0) and WSL 2.7.12. A clean build exposed two genuine reproducibility defects before acceptance: a registry timeout could discard an incomplete pnpm install, and Next.js standalone output omitted a pnpm-linked SWC helper at runtime. The web Dockerfile was hardened with a persistent BuildKit package cache, bounded fetch retries and the resolved dependency tree in the runner image. The rebuilt stack then passed: /v1/health returned ok/synthetic-prototype/0.4.0; /board returned HTTP 200; the complete eight-state Chromium journey reported zero console and page errors; and both containers recovered after Compose restart. Five audit events and the newest SHA-256 event hash were preserved exactly across restart, chain verification remained intact, and explicit reset restored one intact baseline event. This converts the former environmental condition into a passed executable release gate rather than a static configuration claim.

14.3 Vercel deployment architecture and live acceptance

The UI has two explicit execution modes. In local and Docker execution, the Next.js same-origin route proxies to the live FastAPI service through TRIAGELoop_API_URL; this is the reference path for scoring, recomputation, persistent audit state and manual synthetic intake. On Vercel, the same route switches to a bounded presentation adapter. The adapter serves canonical snapshots exported from the FastAPI prototype, uses an HTTP-only, same-site session cookie for small synthetic scenario flags, and appends a deterministic reasoned-decision audit event. It accepts no real patient data and does not claim to be the Python decision service.

This separation prevents two opposite errors: a jury page that renders but has no working interactions, and a hosted mock that is misrepresented as the full clinical engine. The public adapter is evidence-preserving presentation infrastructure; the Docker path remains the technical reference implementation. The environment-aware Next.js configuration retains standalone output for Docker but delegates function tracing to Vercel.

Production deployment dpl_Engkgw53P3J8XbonHm2bWbFb1vY1 reached READY and was aliased to <https://triageloop-ai.vercel.app>. An automated Chromium run against the public origin verified health, reset, baseline board, deterioration with negative Clinical Slack, the NBO fallback boundary, 3x surge, a reasoned override, recomputed audit-chain integrity, evaluation evidence and capacity truth. It captured eight full-page screenshots and reported zero console or page errors. The health endpoint explicitly identifies synthetic-presentation-adapter; production clinical use remains prohibited.

15 Limitations and Threats to Validity

15.1 Construct validity

The generated escalation composite combines several urgent actions. Queue conclusions depend on the response-window definition, as Figure 8 demonstrates. Action Window misses are operational proxy outcomes, not patient harm. The simulator's service and arrival distributions are assumptions rather than measured local operations.

15.2 Internal validity

The project authors designed the generator, labels, features, model, and simulator. A model can learn the generator's own mechanisms. Patient-level isolation, untouched test data, predeclared gates, stress

cohorts, and deterministic reproduction reduce leakage and selection bias but do not remove authoring circularity. The external demo checks scale only.

15.3 External validity

No real ED cohort was used for model performance. Pediatric external validation is absent. The MIMIC-IV demo is adult hospital/ICU data. Site profiles represent configured archetypes, not calibrated hospitals. Thresholds, rules, model calibration, subgroup behavior, workflows, and alert burden require prospective local studies.

15.4 Human factors

The interface has been structurally inspected and browser-tested, but no formal study with triage nurses or assistive-technology users has been completed. Alert workload is measured in simulation without an accepted safe threshold. Typical NBO acquisition time is catalogue-based, not a time-and-motion result.

15.5 Security and regulatory readiness

The SHA-256 chain is tamper-evident but stored locally and is not immutable. The role selector is not authentication. The prototype lacks production authorization, TLS termination, enterprise key management, retention enforcement, monitoring, backup, disaster recovery, penetration testing, and hospital integration approval. Governance references are a design baseline, not legal advice.

15.6 Clinical-safety boundary

The system is not a medical device, is not clinically validated, and must not be used for diagnosis, treatment, discharge, autonomous prioritization, or staffing decisions. Its rules are demonstrations requiring clinical-owner review. Negative Slack identifies an assumed capacity conflict; it does not prescribe the operational response.

16 Prospective Validation and Product Roadmap

The next credible step is not a larger synthetic headline. It is staged local validation:

1. **Silent retrospective validation:** map local ED data, verify units/provenance, measure calibration and subgroup performance without changing care.
2. **Shadow-mode prospective study:** run live predictions invisibly, compare Action Windows and alerts with clinician adjudication, and measure alert burden and override reasons.
3. **Usability and safety simulation:** test time-to-comprehension, error recovery, handoff, surge, outage, and assistive technology with nurses and charge nurses.
4. **Locally governed pilot:** expose recommendations under approved rules, monitoring, identity, audit, escalation ownership, and stop criteria.
5. **Outcomes evaluation:** only after calibration and workflow safety are established, evaluate process and patient outcomes with an appropriate controlled design.

Priority engineering work includes audited correction of source data, site-specific rule/configuration governance, production identity and authorization, monitored event storage, integration adapters, formal SSE/reconnect semantics, local recalibration, drift monitoring, and an NBO redesign that must pass an independent stress gate before any efficiency claim.

17 Conclusion

TriageLoop reframes emergency-department waiting-room triage from a one-time score into a patient-specific, capacity-aware, clinician-closed control loop. Its key design move is to translate changing clinical evidence into a conservative Action Window, keep queue feasibility separate through Clinical Slack, and make uncertainty produce a specific safe workflow rather than a decorative confidence percentage.

The synthetic evidence supports internal consistency and a positive operational hypothesis under declared assumptions. It also rejects the single-observation NBO hypothesis and reveals dependence on site capacity and response definition. Those negative and conditional results are part of the contribution. The prototype's value at this stage is not proof of clinical benefit; it is a rigorously bounded, reproducible demonstration of how safety rules, calibrated trajectory risk, uncertainty, queue feasibility, human authority, and audit can operate as one system.

A Clinical Calculation Pseudocode

```

validate(patient_state, provenance, freshness, plausibility)
safety = evaluate_age_aware_rules(patient_state)

if safety.hard_red_flag:
    action = IMMEDIATE_REVIEW
    window = 0 minutes
else:
    risk = calibrated_multi_horizon_hazard(features_at_or_before_now)
    conformal_set = mondrian_prediction_set(risk)
    shifted = ood_detector(features)
    safe_mode = safety.data_quality_failure or shifted or material_ambiguity

    model_window = earliest_horizon_crossing(risk, thresholds)
    category_window = max(0, category_bound - elapsed_wait)
    window = min(model_window, category_window, safe_mode_bound)
    action = conservative_action(window, safety, clinician_state)

eta = queue_twin.projected_time_to_action(patient, current_queue, resources)
slack = window - eta
nbo = first_step_observation_only_if_decision_relevant_uncertainty

present(action, window, eta, slack, reasons, uncertainty, nbo)
clinician_decision = accept_or_modify_or_override(reason_if_required)
append_and_recompute_hash_linked_audit(clinician_decision)

```

B API Surface

Endpoint	Method	Purpose
/v1/health	GET	Health, prototype mode, and API version
/v1/config	GET	Site/scenario configuration and safety boundaries
/v1/demo/state	GET	Complete deterministic board state
/v1/demo/reset	POST	Restore baseline fixtures and audit genesis
/v1/demo/scenario	POST	Select baseline or 3x surge
/v1/demo/deteriorate/{id}	POST	Apply registered repeated-observation hero event
/v1/intake/manual	POST	Strict synthetic manual intake
/v1/patients/{id}/recommendation	GET	Recommendation envelope with risk, uncertainty, Window, Slack, NBO, explanation
/v1/patients/{id}/decisions	POST	Accept, modify, or override; reason required where applicable
/v1/audit	GET	Ordered material event history
/v1/audit/integrity	GET	Full recomputation of hash and previous-link integrity
/v1/evaluations/latest	GET	Machine-readable model, queue, NBO, and plausibility evidence
/v1/events/stream	GET	Server-sent state/heartbeat stream

C Claim-Control Ledger

Permitted claim	Required qualifier	Prohibited expansion
20.5% fewer Action Window misses	Synthetic 3x-surge simulation versus fixed 15-minute periodic re-triage; 95% CI 17.4%-23.8%	Patient benefit, mortality reduction, or real-hospital performance
NBO requested 87.5% fewer observations	It lost 7.1 recall points and failed; first-step only	Workload reduction with preserved safety
63,163 external measurements mapped	Open adult hospital/ICU demo; input scale only	External model validation or ED transfer
Critical recall/coverage gates pass	Generated test/stress cohorts	Clinical accuracy, fairness, or regulatory performance
Hash chain detects tampering	Local prototype recomputation	Immutable, certified, or non-repudiable audit
Structural accessibility checks pass	Browser/DOM and stylesheet inspection	Formal WCAG conformance or nurse usability
India governance baseline used	Design framing only	DPDP/ABDM legal compliance or certification

D Artifact and Evidence Map

Artifact	Role in this monograph
docs/clinical-safety-contract.md	Safety invariants, authority, failure and audit boundaries
docs/evaluation-protocol.md	Predeclared research questions, endpoints, gates, model-selection rule
docs/model-card.md	Data isolation, selected-model results, uncertainty and limitations
docs/simulation-card.md	Site/load assumptions, policy definitions, queue evidence and limits
docs/requirements-traceability.md	Official-brief requirement-to-evidence mapping
docs/tl-04.5-critique-disposition.md	First independent-review verification and corrective actions
docs/tl-06.5-final-review-disposition.md	Final panel finding-by-finding resolution
docs/tl-05-verification-report.md	Scientific, product, security, accessibility, and reproducibility checkpoint
docs/tl-07-release-report.md	Final local full-story, test/build, video, and release-package evidence
artifacts/release/tl07-docker-verification.json	Docker versions, endpoint/browser results, restart persistence, reset contract, and fixes found by the gate

Artifact	Role in this monograph
paper/diagrams/*.mmd	Editable Mermaid sources for the system architecture, runtime clinical-decision flow, and ML-development pipeline blueprints
artifacts/evaluation/tl-02-metrics.json	Candidate/selected model metrics and subgroup evidence
artifacts/evaluation/tl-03-queue-metrics.json	Original 1,800-shift matrix and static comparator
artifacts/evaluation/tl-05-periodic-retriage-metrics.json	Stronger 1,200-shift comparator, sensitivity, and workload
artifacts/evaluation/tl-05-nbo-metrics.json	Failed NBO gate and fallback sensitivity
artifacts/evaluation/tl-05-external-plausibility.json	Open-reference input-scale checks
submission/visuals/product-surfaces/	Six direct working-product evidence captures

E Reproduction Checklist

1. Confirm Python 3.12+, Node 22+, pnpm, and optional Docker Desktop.
2. Rebuild generated encounters with seed 20260821; confirm snapshot counts and manifest.
3. Retrain candidates; confirm selected model family and model SHA-256.
4. Rerun original queue, periodic comparator, NBO, and external-plausibility scripts.
5. Compare evaluation hashes with the recorded values.
6. Run all 73 Python tests, TypeScript check, and optimized Next.js build.
7. Start API/web; execute reset, deterioration, surge, override, audit, evidence, capacity, and outage/recovery.
8. With Docker, follow the README Compose steps; verify health, board, restart/persistence, and reset.
9. For Vercel, connect the API and restrict CORS. Verify the synthetic workflow from the public deployment.
10. Recheck all external links and retain the synthetic/non-clinical boundary on every public artifact.

References

- [1] Accenture Innovation Challenge 2026. *Round 2 - Prototype Development: PatientTriage.ai*. Supplied competition brief, pp. 2-4, 2026.
- [2] V. M. Quinten, M. van Meurs, T. J. Olgers, J. M. Vonk, J. J. M. Ligtenberg, and J. C. ter Maaten. "Repeated vital sign measurements in the emergency department predict patient deterioration within 72 hours: a prospective observational study." *Scandinavian Journal of Trauma, Resuscitation and Emergency Medicine*, 26:57, 2018. doi:10.1186/s13049-018-0525-y.
- [3] I. J. Brekke et al. "The value of vital sign trends in predicting and monitoring clinical deterioration: A systematic review." *PLOS ONE*, 14(1), 2019. PMID 30645637.
- [4] "A High-Precision Time-Varying Survival Model for Early Prediction of Patient Deterioration: A Retrospective Cohort Study." PubMed PMID 41827109, 2026. <https://pubmed.ncbi.nlm.nih.gov/41827109/>.
- [5] "Development of an Anticipatory Triage-Ranking Algorithm Using Dynamic Simulation of the Expected Time Course of Patients With Trauma: Modeling and Simulation Study." PubMed PMID 37318826, 2023. <https://pubmed.ncbi.nlm.nih.gov/37318826/>.
- [6] A. N. Angelopoulos and S. Bates. "Conformal Prediction: A Gentle Introduction." *Foundations and Trends in Machine Learning*, 16(4):494-591, 2023. doi:10.1561/2200000101.
- [7] A. S. B. Abdulai, J. Storm, and M. Ehrlich. "I don't know: An uncertainty-aware machine learning model for predicting patient disposition at emergency department triage." *International Journal of Medical Informatics*, 201:105957, 2025. doi:10.1016/j.ijmedinf.2025.105957.
- [8] NASA. *Prognostics and Remaining Useful Life research reference*. NASA Technical Reports Server, 2014. <https://ntrs.nasa.gov/api/citations/20140010623/downloads/20140010623.pdf>.
- [9] C. L. Liu and J. W. Layland. "Scheduling Algorithms for Multiprogramming in a Hard-Real-Time Environment." *Journal of the ACM*, 20(1):46-61, 1973.
- [10] IBM Research. *Active Feature Acquisition*. Research project overview. <https://research.ibm.com/projects/active-feature-acquisition>. Accessed 22 August 2026.
- [11] World Health Organization. *Interagency Integrated Triage Tool*. <https://www.who.int/tools/triage>. Accessed 22 August 2026.
- [12] R. Gupta, A. Kumar, and S. Dang. "Agentic AI for Clinical Urgency Mapping and Queue Optimization in High-Volume Outpatient Departments: A Simulation-Based Evaluation." arXiv:2604.00215, 2026. <https://arxiv.org/abs/2604.00215>.
- [13] PhysioNet. *MIMIC-IV Clinical Database Demo, version 2.2*. DOI 10.13026/dp1f-ex47. <https://physionet.org/content/mimic-iv-demo/2.2/>.
- [14] PhysioNet. *MIMIC-IV-ED, version 2.2*. Credentialed data resource. <https://physionet.org/content/mimic-iv-ed/2.2/>.
- [15] Government of India. *Digital Personal Data Protection Act, 2023*. India Code. <https://www.indiacode.nic.in/handle/123456789/22037>.
- [16] Ministry of Electronics and Information Technology. *Digital Personal Data Protection Rules, 2025*. Government of India. <https://www.meity.gov.in/>.
- [17] National Health Authority. *Ayushman Bharat Digital Mission Health Data Management Policy*. <https://abdm.gov.in/>.
- [18] Ministry of Health and Family Welfare. *Electronic Health Record Standards for India*. Government of India, 2016. <https://www.mohfw.gov.in/>.

Final boundary: TriageLoop is a synthetic research and product prototype. It does not autonomously diagnose, treat, discharge, downgrade, determine staffing, or replace licensed clinical judgment.